

$$KE_{\text{(linear)}} \text{ of } \Delta m_i = \frac{1}{2} \Delta m_i v_i^2 = \frac{1}{2} \Delta m_i (\omega r_i)^2 = \frac{1}{2} \Delta m_i \omega^2 r_i^2 \quad (KE_{\text{(rot)}} \text{ of } \Delta m_i)$$

$$\therefore \text{Total } KE_{\text{(rotation)}} = \sum_{i=1}^n \frac{1}{2} \Delta m_i \omega^2 r_i^2 = \frac{1}{2} \omega^2 \sum_{i=1}^n \Delta m_i r_i^2$$

$$\text{but } K = \frac{1}{2} I \omega^2 \quad \therefore I = \sum_{i=1}^n \Delta m_i r_i^2 = \int r^2 dm$$

- (d) 'n Gedeelte van 'n dun uniforme staaf het 'n massa dm en sy lengte is $d\ell$. Skryf 'n uitdrukking neer vir die lyndigtheid λ van $d\ell$. Gee ook die eenheid van λ .

A part of a uniform thin rod has a mass dm and its length is $d\ell$. Give an expression for the line density λ of $d\ell$. Also give the unit for λ .

$$\lambda = \frac{dm}{d\ell} \quad (\text{Unit} = \frac{\text{kg}}{\text{m}} = \text{kg} \cdot \text{m}^{-1})$$

- (e) Beskou 'n dun uniforme staaf met massa M en lengte L . Toon aan die traagheidsmoment om 'n as loodreg op die staaf en deur sy swaartepunt gegee word deur $I = \frac{1}{12} ML^2$.

A thin uniform rod has a mass M and length L . Show that the moment of inertia about an axis which is perpendicular to the rod and which goes through its center of mass is given by $I = \frac{1}{12} ML^2$.

Divide into small parts dm $\therefore \lambda = \frac{dm}{d\ell}$ $dm = \lambda d\ell$

$$\text{but } \lambda = \frac{M}{L} \quad \therefore dm = \frac{M}{L} d\ell$$

$$I = \int x^2 dm = \int_{-L/2}^{L/2} x^2 \frac{M}{L} d\ell$$

$$I = \frac{M}{L} \int_{-L/2}^{L/2} x^2 d\ell = \frac{M}{L} \left[\frac{1}{3} x^3 \right]_{-L/2}^{L/2}$$

$$= \frac{M}{3L} \left(\frac{L^3}{8} + \frac{L^3}{8} \right) = \frac{M}{3L} \left(\frac{L^3}{4} \right) = \frac{1}{12} ML^2$$

- (f) Wanneer die rotasie-as steeds loodreg op bogenoemde staaf is, maar deur die staaf se een eindpunt gaan, bepaal die staaf se traagheidsmoment om hierdie as deur gebruik te maak van die teorie van parallelle asse.

With the axis of rotation is still perpendicular to the rod mentioned above, but it goes through its one end, determine the moment of inertia about this axis by applying the parallel-axis theorem.

$$I = I_{\text{cm}} + Md^2 = \frac{1}{12} ML^2 + M \left(\frac{L}{2} \right)^2$$

$$= \frac{1}{12} ML^2 + \frac{1}{4} ML^2$$

$$= \frac{1}{3} ML^2$$